King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 101 Major Exam I Term 233

Original exam date: 10 July 2024. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. If the function $g(x) = \begin{cases} \dfrac{4\sin x}{x}, & x < 0 \\ a - 2x, & x \geq 0 \end{cases}$ is continuous at $x = 0$, then $a =$

(a) 2

(b) 0

(c) -4

(d) 4

(e) -2

2. $\lim_{x\to 16} \dfrac{4 - \sqrt{x}}{x - 16} =$

(a) $\dfrac{1}{8}$

(b) $\dfrac{-1}{8}$

(c) $\dfrac{1}{2}$

(d) $\dfrac{1}{4}$

(e) $\dfrac{-1}{4}$

3. If $y = \dfrac{3}{(2x)^3} + 2\sin x$, then $\dfrac{dy}{dx}$ at $x = \dfrac{\pi}{2}$ is equal to

(a) $\dfrac{-9}{\pi^4}$

(b) $\dfrac{-18}{\pi^4}$

(c) $\dfrac{9}{\pi^4}$

(d) $\dfrac{1}{\pi^4}$

(e) $\dfrac{18}{\pi^4}$

4. If $y = ax + b$ is the equation of the tangent line to the graph of the function $f(x) = \dfrac{1 + \cos x}{1 - \cos x}$ at the point $\left(\dfrac{\pi}{2}, 1\right)$, then $a + b =$

(a) $\pi + 1$

(b) $\pi + 2$

(c) π

(d) $\pi - 1$

(e) $\pi - 2$

5. Which one of the following statements is False?

(a) The graph of every rational function has a vertical asymptote

(b) The graph of a rational function may have several vertical asymptotes

(c) The graphs of polynomial functions have no vertical asymptotes

(d) The graph of a function cannot cross a vertical asymptote

(e) The graphs of trigonometric functions may have vertical asymptotes

6. If the function $f(x) = \dfrac{8(x-2)}{e^x}$ has a horizontal tangent line at $x = a$, then $a =$

(a) 4

(b) 3

(c) 1

(d) 5

(e) 2

7. If the tangent line to the graph of $y = h(x)$ at the point $(-1, 4)$ passes through the point $(3, 6)$, then $h(-1) + 2h'(-1) =$

(a) 3

(b) 5

(c) 7

(d) 6

(e) 4

8. If $h(x) = \dfrac{x^5 + 2x + 6}{x^{1/3}}$, then $h'(1) =$

(a) 3

(b) 4

(c) 5

(d) 6

(e) 2

9. The number of vertical asymptotes of the graph of the function $f(x) = \dfrac{4x^2 + 4x - 24}{x^4 - 2x^3 - 9x^2 + 18x}$ is

(a) 1

(b) 0

(c) 2

(d) 3

(e) 4

10. $\lim_{x \to -1} \left(\left\lfloor \dfrac{x}{3} \right\rfloor + 3 \right)$, where $\lfloor t \rfloor$ denotes the greatest integer function (printed as double brackets in the exam)

(a) equals 2

(b) equals 0

(c) equals 3

(d) equals -2

(e) does not exist

11. If $y = k$ is a horizontal asymptote for the function $f(x) = \dfrac{\sqrt{9x^2 - 2}}{2x + 1}$, then the product of all possible values of k is

(a) $\dfrac{-9}{4}$

(b) $\dfrac{3}{4}$

(c) $\dfrac{-9}{2}$

(d) $\dfrac{9}{2}$

(e) $\dfrac{-3}{4}$

12. $\lim_{x\to\infty} x \tan \dfrac{1}{x}$

(a) equals 0

(b) does not exist

(c) equals 1

(d) equals π

(e) equals -1

13. If $f(x) = (x^3 - x)(x^2 + 2)(x^2 + x - 1)$, then $f'(1) =$

(a) 8

(b) 2

(c) 10

(d) 6

(e) 4

14. $\lim_{x\to 2} \dfrac{x^3 - 8}{x - 2} =$

(a) 10

(b) 12

(c) 8

(d) 14

(e) 16

Answer Key

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Q	Ans	Brief explanation
1	**(d)**	$\lim_{x\to 0^-} \frac{4\sin x}{x} = 4$ and $g(0) = a$, so $a = 4$.
2	**(b)**	Conjugate: $\frac{-1}{4+\sqrt{x}} \to -\frac{1}{8}$.
3	**(b)**	$y = \frac{3}{8}x^{-3} + 2\sin x$, $y' = -\frac{9}{8}x^{-4} + 2\cos x$, giving $-\frac{18}{\pi^4}$.
4	**(d)**	$f'(\frac{\pi}{2}) = -2$, tangent $y = -2x + \pi + 1$, so $a + b = \pi - 1$.
5	**(a)**	A polynomial (or $\frac{1}{x^2+1}$) is rational with no vertical asymptote, so the first statement is false.
6	**(b)**	$f'(x) = \frac{8(3-x)}{e^x} = 0 \Rightarrow a = 3$.
7	**(b)**	$h(-1) = 4$, $h'(-1) = \frac{6-4}{3+1} = \frac{1}{2}$, so the value is 5.
8	**(b)**	$h = x^{14/3} + 2x^{2/3} + 6x^{-1/3}$; $h'(1) = \frac{14}{3} + \frac{4}{3} - 2 = 4$.
9	**(c)**	$f = \frac{4}{x(x-3)}$ after cancelling; asymptotes at $x = 0$ and $x = 3$: 2.
10	**(a)**	Near $x = -1$, $\lfloor x/3 \rfloor = -1$ (since $x/3 \approx -\frac{1}{3}$); limit $= -1 + 3 = 2$.
11	**(a)**	Limits at $\pm\infty$ are $\pm\frac{3}{2}$, so the product is $-\frac{9}{4}$.
12	**(c)**	$t = 1/x$: $\frac{\tan t}{t} = \frac{\sin t}{t} \cdot \frac{1}{\cos t} \to 1$.
13	**(d)**	$u = x^3 - x$ vanishes at 1: $f'(1) = u'(1)v(1)w(1) = 2 \cdot 3 \cdot 1 = 6$.
14	**(b)**	$x^3 - 8 = (x-2)(x^2+2x+4)$, limit $= 12$.